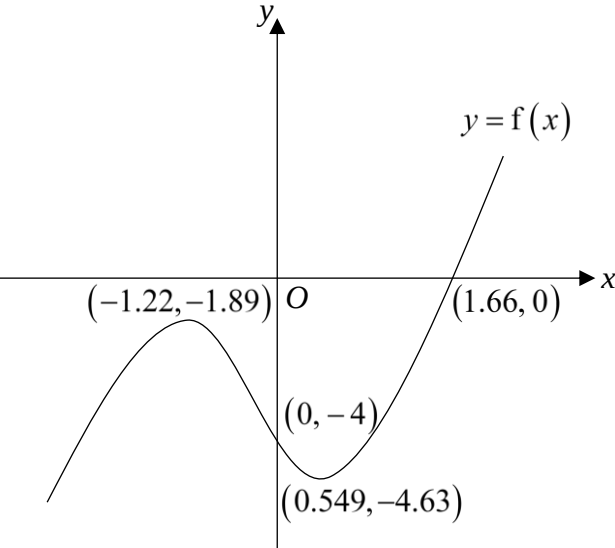


2012 'A' Level H2 Mathematics Paper 2 Suggested Solutions

Section A: Pure Mathematics [40 marks]

Qn 1	Suggested Solutions
(a)	$\frac{d^2y}{dx^2} = 16 - 9x^2$ $\Rightarrow \int \frac{d^2y}{dx^2} dx = \int (16 - 9x^2) dx$ $\Rightarrow \frac{dy}{dx} = 16x - 3x^3 + A$ $\Rightarrow \int \frac{dy}{dx} dx = \int (16x - 3x^3 + A) dx$ $\Rightarrow y = 8x^2 - \frac{3}{4}x^4 + Ax + B$ Integrate twice to get equation of y in terms of x.
(b)	$\frac{du}{dt} = 16 - 9u^2$ $\Rightarrow \int \frac{du}{16 - 9u^2} = \int 1 dt \leftarrow$ Use separable variables method to solve DE. $\Rightarrow \frac{1}{24} \ln \left \frac{4+3u}{4-3u} \right = t + B$ $\Rightarrow \ln \left \frac{4+3u}{4-3u} \right = 24t + 24B$ Always remove the modulus before you substitute in the conditions to find your arbitrary constants. $\Rightarrow \frac{4+3u}{4-3u} = \pm e^{24t+24B}$ $\Rightarrow \frac{4+3u}{4-3u} = Ae^{24t}, A = \pm e^{24B}$ When $t = 0, u = 1, A = 7$ $\therefore \frac{4+3u}{4-3u} = 7e^{24t}$ $\Rightarrow e^{24t} = \frac{4+3u}{7(4-3u)}$ $\therefore t = \frac{1}{24} \ln \left(\frac{4+3u}{7(4-3u)} \right)$

Qn 2 Out of syllabus (9758)

Qn 3	Suggested Solutions
(i)	 Label all relevant features of the graph. For example, x- and y-intercepts and turning points (to 3 s.f.).
(ii)	$f(x) = 4$ $x^3 + x^2 - 2x - 4 = 4$ $x^3 + x^2 - 2x - 8 = 0$ Using GC, $x = 2$. $x^3 + x^2 - 2x - 8 = (x - 2)(x^2 + ax + b)$ Comparing coefficients of, x^2: $a - 2 = 1 \Rightarrow a = 3$ constant: $-2b = -8 \Rightarrow b = 4$ $\therefore x^3 + x^2 - 2x - 8 = (x - 2)(x^2 + 3x + 4)$ $(x - 2)(x^2 + 3x + 4) = 0$ $x = 2$ or $x^2 + 3x + 4 = 0$ $\left(x - \frac{3}{2}\right)^2 \geq 0$ for all real values of x However, $x^2 + 3x + 4 = \left(x - \frac{3}{2}\right)^2 + \frac{7}{4} > 0$ for all real values of x. Hence there are no other real solutions (proved). <u>Alternatively,</u> Discriminant of $x^2 + 3x + 4$ is $(3)^2 - 4(1)(4) = -7 < 0$ and coefficient of x^2 is positive. $\therefore x^2 + 3x + 4 > 0$ for all real values of x. Hence there are no other real solutions (proved).
(iii)	Replace x by $x + 3$ From part (ii), $x + 3 = 2 \Rightarrow x = -1$

(iv)

(v)

When $|f(x)| = 4$,

$$x^3 + x^2 - 2x - 4 = 4 \quad \text{or} \quad -(x^3 + x^2 - 2x - 4) = 4$$

Therefore, the required cubic equations are:

$$x^3 + x^2 - 2x - 8 = 0 \quad \text{or} \quad x^3 + x^2 - 2x = 0$$

For $x^3 + x^2 - 2x - 8 = 0 \Rightarrow x = 2$ (from part (ii))

For $x^3 + x^2 - 2x = 0$,

Using GC, $x = 0$ or $x = 1$ or $x = -2$

$\therefore$ roots of $|f(x)| = 4$ are $-2, 0, 1$ and 2 .

$|f(x)| = 4 \Rightarrow f(x) = 4 \quad \text{or} \quad -f(x) = 4$

Cubic equations are of the form $ax^3 + bx^2 + cx + d = 0$.

Qn 4 Suggested Solutions

(i) Let n be the number of months from January 2001.
Amount of money in account on first day of n th month in dollars,

$$S_n = 100 + 110 + 120 + \cdots + [100 + (n-1)10]$$

$$= \frac{n}{2}[2(100) + (n-1)10]$$

$$S_n > 5000$$

$$\frac{n}{2}[2(100) + (n-1)10] > 5000$$
Using GC,
when $n = 23$, $S_{23} = 4830 < 5000$,
when $n = 24$, $S_{24} = 5160 > 5000$.
Minimum $n = 24$
Mrs A's account became greater than \$5000 on 1 Dec 2002.

Amount in account **increases by a constant amount** every month, hence use sum of an arithmetic progression with $a = 100$ and $d = 10$.

Find the **first** value of n such that $S_n > 5000$ (show the sign change).

(ii)

Month	Start	End
1	100	$100(1.005)$
2	$100(1.005) + 100$	$100(1.005)^2 + 100(1.005)$
3	$100(1.005)^2 + 100(1.005) + 100$	$100(1.005)^3 + 100(1.005)^2 + 100(1.005)$
...		
n		$100(1.005)^n + 100(1.005)^{n-1} + \cdots + 100(1.005)$

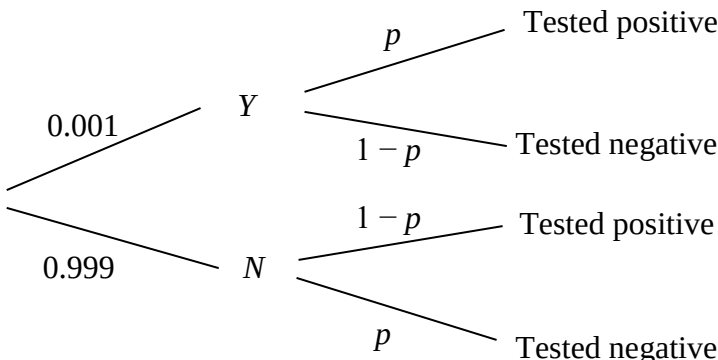
Amount at the end of n th month

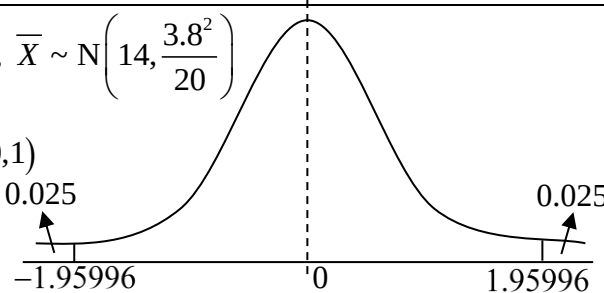
Use a table to spot patterns. In this case, sum of a geometric progression is used with $a = 100$ and $r = 1.005$.

	$= 100(1.005)^n + 100(1.005)^{n-1} + \dots + 100(1.005)$ $= \frac{100(1.005)}{1-1.005}(1-1.005^n)$ $= 20100(1.005^n - 1)$ $20100(1.005^n - 1) > 5000$ $\ln[20100(1.005^n - 1)] > \ln 5000$ $\Rightarrow n > \frac{\ln\left(\frac{5000}{20100} + 1\right)}{\ln(1.005)}$ $\Rightarrow n > 44.54$ Mr B's account became greater than \$5000 at the end of the 45th month, i.e. September 2004. Alternatively, using GC, when $n = 44$, $20100(1.005^n - 1) = 4932.42 < 5000$, when $n = 45$, $20100(1.005^n - 1) = 5057.58 > 5000$. Minimum $n = 45$ Mr B's account became greater than \$5000 at the end of the 45th month, i.e. September 2004.
(iii)	Let $y\%$ be the interest rate per month. Let $x = 1 + \frac{y}{100}$. $100x\left(\frac{1-x^{35}}{1-x}\right) + 100 = 5000$ $100x\left(\frac{1-x^{35}}{1-x}\right) - 4900 = 0$ $x\left(\frac{1-x^{35}}{1-x}\right) - 49 = 0$ Using GC, $x = 1.017957 \Rightarrow y = 1.7957$. The interest rate is 1.80% per month. Use method in part (ii) to solve for the interest rate, replacing original interest of 0.5% with $y\%$. Amount in account after 35 months $= 100x^{35-1} + 100x^{35-2} + 100x^{35-3} + \dots + 100x$ $= 100x\left(\frac{1-x^{35}}{1-x}\right)$ Note: Mr B will add \$100 on 1 December 2003 but interest will not be applied until the end of the month (i.e. do not consider interest to be applied in December 2003). Equate to 5000 since Mr B wanted the amount in the account to be \$5000 on 1 December 2003

Section B: Statistics [60 Marks]

Qn 5	Suggested Solutions
	Let Y denotes the event "the patient has disease" Let N denotes the event "the patient does not have disease"

Qn 5	Suggested Solutions
	
(i) (a)	Given that $p = 0.995$ $P(\text{test result is positive})$ $= (0.001)(0.995) + (0.999)(0.005)$ $= 0.00599$ Check that the tree diagram is correct before answering questions. Also, always check that the correct branches are used in calculations.
(i) (b)	$P(\text{has disease} \mid \text{test result is positive})$ $= \frac{P(\text{has disease} \cap \text{test result is positive})}{P(\text{test result is positive})}$ $= \frac{(0.001)(0.995)}{0.00599} = 0.166 \quad (3 \text{ s.f.})$
(ii)	$P(\text{has disease} \mid \text{test result is positive}) = 0.75$ $\Rightarrow \frac{0.001p}{0.001p + 0.999(1-p)} = 0.75$ Use method in part (i)(b) but use unknown p instead of $p = 0.995$. $0.001p = 0.00075p + 0.74925(1-p)$ $0.7495p = 0.74925$ $p = 0.999666 \quad (6 \text{ d.p.})$

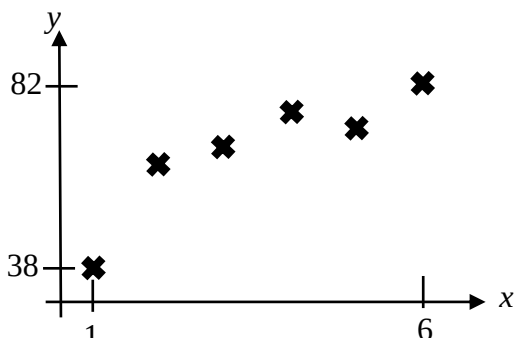
Qn 6	Suggested Solutions
(i)	Let X be the tail lengths of a randomly chosen squirrel Let μ denote the population mean tail lengths of a squirrel $H_0: \mu = 14$ $H_1: \mu \neq 14$
(ii)	Under H_0, since $X \sim N(14, 3.8^2)$, $\bar{X} \sim N\left(14, \frac{3.8^2}{20}\right)$ Test Statistic: $Z = \frac{\bar{X} - 14}{\frac{3.8}{\sqrt{20}}} \sim N(0, 1)$ Level of significance: 5% Reject H_0 if $p\text{-value} < 0.05$ i.e. For H_0 not to be rejected, $-1.95996 < z\text{-value} < 1.95996$ 

Check which test (left-tailed, right-tailed or two-tailed) is used before using invNorm.

For two-tailed test, recall that you need to consider that the level of significance is halved for each end (see diagram).

Qn 6	Suggested Solutions
	$-1.95996 < \frac{\bar{x} - 14}{\frac{3.8}{\sqrt{20}}} < 1.95996$ $-1.6654 < \bar{x} - 14 < 1.6654$ $12.3 < \bar{x} < 15.7 \quad (3 \text{ s.f.})$ Set of values of $\bar{x}$ is $\{\bar{x} \in \mathbb{R} : 12.3 < \bar{x} < 15.7\}$
(iii)	If $\bar{x} = 15.8$, we reject H_0 and conclude that there is sufficient evidence at the 5% level of significance to conclude that the squirrels on the island do not have the same tail length as the species known to her. Note: Must answer in context of the question.

Qn 7	Suggested Solutions
(i)	P (the sisters are next to each other) $= \frac{14! \times 2!}{15!} = \frac{2}{15}$ Sisters are grouped together as one unit, then permuted among themselves.
(ii)	P (the brothers are not all next to each other) $= 1 - P(\text{the brothers are all next to each other})$ $= 1 - \frac{13! \times 3!}{15!} = \frac{34}{35}$ Brothers are grouped together as one unit, then permuted among themselves.
(iii)	P (the sisters are next to each other and the brothers are all next to one another) $= \frac{12! \times 2! \times 3!}{15!} = \frac{2}{455}$ Sisters are grouped together as one unit, brothers are grouped together as one unit, then permuted within their groups.
(iv)	Let event A: the sisters are next to each other Let event B: the brothers are all next to one another $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $= \frac{2}{15} + \left(1 - \frac{34}{35}\right) - \frac{2}{455}$ $= \frac{43}{273}$ Important formula: use a Venn diagram (if needed) to recall.
(v)	P (the sisters are next to each other) $= \frac{(14-1)! \times 2!}{(15-1)!} = \frac{1}{7}$ Similar to part (i) in method but must consider arrangements in a circle.

Qn 8	Suggested Solutions																																													
(i)		Remember to indicate coordinates of first and last points on axes.																																												
(ii)	The irregularity occurred in Week 1. That practice paper may be more difficult than the other papers. Note: Another possible reason could be that Amy was not prepared academically for the practice paper in Week 1.																																													
(iii)	The marks cannot exceed 100%, and so a linear model, which models an infinite upward/downward trend of data, is not appropriate. The marks are likely to plateau off or stay constant as the weeks go by, rather than in the case of a quadratic model which is expected to fit data with an increase and then a decrease (or the other way round) trend. Thus, a quadratic model is also not appropriate.																																													
(iv)	For $L = 91$, $r = -0.929744$ (6 decimal places) <table border="1" data-bbox="314 1046 557 1274"><thead><tr><th>L₁</th><th>L₂</th><th>L₃</th><th>θ</th></tr></thead><tbody><tr><td>1</td><td>38</td><td>3.9703</td><td></td></tr><tr><td>2</td><td>63</td><td>3.3322</td><td></td></tr><tr><td>3</td><td>67</td><td>3.1781</td><td></td></tr><tr><td>4</td><td>75</td><td>2.7726</td><td></td></tr><tr><td>5</td><td>71</td><td>2.9957</td><td></td></tr><tr><td>6</td><td>82</td><td>2.1972</td><td></td></tr><tr><td>-----</td><td>-----</td><td>-----</td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr></tbody></table> L₃ = "ln(91-L₂)" LinReg(a+bx) Xlist:L₁ Ylist:L₃ FreqList: Store RegEQ: Calculate LinReg y=a+bx a=4.102371154 b=-0.2937205285 r²=0.8644241249 r=-0.929744118		L ₁	L ₂	L ₃	θ	1	38	3.9703		2	63	3.3322		3	67	3.1781		4	75	2.7726		5	71	2.9957		6	82	2.1972		-----	-----	-----													
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(v)	Since $r = 0.929944$ is closest to 1 for $L = 92$, this is the most appropriate value for L.																																													
(vi)	From GC, $a = 4.1045 = 4.10$, $b = -0.27960 = -2.80$ (to 3 s.f.) Thus, $\ln(92 - 90) = 4.1045 - 0.27960x$ $\Rightarrow x = 12.2$ Amy will get at least 90% in Week 13. Value is rounded up, since in Week 12, Amy would not have achieved at least 90% yet.																																													
(vii)	L is the percentage mark she gets if she continues practising indefinitely. Recall that $\ln(92 - y) = 4.1045 - 0.27960x \rightarrow -\infty$ as $x \rightarrow \infty$, hence $y \rightarrow 92$ as $\ln(92 - y) \rightarrow -\infty$.																																													

Qn 9	Suggested Solutions
(i)	The probability that a randomly chosen voter supports the Alliance Party is constant for all the voters. Whether a randomly chosen voter supports the Alliance Party is independent of any other voter.

Identify the **trial** (voter) and the **success** (voter supports the Alliance Party) first. Then check **FIST**.

(ii)	$P(A = 3 \text{ or } 4)$ $= P(A = 3) + P(A = 4)$ $= 0.37307 = 0.373 \quad (3 \text{ s.f.})$
(iii)	Out of syllabus (9758)
(iv)	$P(A = 15) = 0.06864$ $\binom{30}{15} p^{15} (1-p)^{15} = 0.06864 \quad \leftarrow \text{Use formula found in MF26.}$ $[p(1-p)]^{15} = \frac{0.06864}{\binom{30}{15}}$ $\therefore p(1-p) = 0.23790 \quad (\text{shown})$ $k = 0.238 \quad (3 \text{ s.f.}) \quad \leftarrow \text{State } k \text{ since question says “} k \text{ is a constant to be determined”}.$ Using GC, $p = 0.390 \quad \text{or} \quad p = 0.610 \quad (3 \text{ s.f.})$ Since $p < 0.5$, $p = 0.390 \quad (3 \text{ s.f.})$ Use the 5 s.f. value of k to find the value of p. Degree of accuracy should be at least 3 s.f.

Qn 10 Out of syllabus (9758)